

DANYAL TANVEER

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SUMMARY

Computer Science graduate (BS, University of Central Punjab, CGPA 3.60/4.00) specializing in Artificial Intelligence and Machine Learning, with hands-on experience in transformer fine-tuning (RoBERTa, DeBERTa, ELECTRA, BERT, DistilBERT), Retrieval-Augmented Generation (RAG) pipelines, NLP, and LLM-based tooling. Proficient in Python, PyTorch, TensorFlow, and HuggingFace Transformers, with a strong foundation in data structures, algorithms, and both SQL and NoSQL databases. Co-author of a fake-news-detection research preprint and builder of production-grade AI systems, including a capstone news-credibility platform processing 220K+ labeled samples. Combines applied ML depth with full-stack engineering experience shipping real-world software.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML/CSS
AI and ML: PyTorch, TensorFlow, HuggingFace Transformers, Transformer Fine-Tuning (BERT, RoBERTa, ELECTRA, DistilBERT, DeBERTa), RAG Pipelines (chunking, embeddings, vector search), NLP, LLMs & AI Agents, scikit-learn, NumPy, Pandas, Gradio
Frameworks: Django, DRF, FastAPI, React, React Native, Next.js, Node.js, Express
Databases: MongoDB, MySQL, PostgreSQL + pgvector (vector search), Firebase, SQLite
Tools and DevOps: Git/GitHub, CI/CD, Postman, Swagger/OpenAPI, JWT Auth, Redis, Jupyter Notebook, Agile/Scrum

CAPSTONE PROJECT

TRAK: AI-Powered News Credibility Platform Jan 2026 – Present
Capstone Project, University of Central Punjab | React Native, Django, MongoDB, HuggingFace Transformers

- Architected a full-stack mobile application using React Native (frontend) and Django/DRF (backend) with MongoDB for persistent data storage, applying OOP design patterns throughout.
- Fine-tuned RoBERTa and DeBERTa transformer models on 220K+ labeled samples for three-class credibility classification: real, fake, and suspicious.
- Built an end-to-end automated pipeline covering news scraping, NLP preprocessing, credibility classification, and personalized user feed delivery.
- Implemented JWT authentication, role-based access control, text summarization, and text-to-speech features.

Tech: React Native, Django, DRF, MongoDB, Python, HuggingFace Transformers, JWT

RESEARCH

Multi-Transformer Sentimental LIAR: Fake News Detection Research Mar 2026 – Present
University of Central Punjab | Preprint submitted to Elsevier

- Co-authored a preprint benchmarking matched RoBERTa, ELECTRA, BERT, and DistilBERT transformers on the LIAR short-claim dataset, reporting Accuracy, Precision, Recall, F1, AUC, and MCC.
- Achieved 72.03% accuracy (F1 0.7626, MCC 0.4247) with ELECTRA on the primary text+sentiment+emotion setting; ran systematic ablations across text-only, sentiment, emotion, and combined feature paths.
- Engineered an affective feature pipeline using VADER sentiment and NRC emotion lexicons concatenated with claim text for transformer fine-tuning; deployed an interactive Gradio demo using DistilBERT for efficient inference.

Tech: Python, HuggingFace Transformers, PyTorch, scikit-learn, VADER, NRCLex, Gradio, Pandas, NumPy

Throne of Glass Archive: Retrieval-Augmented Generation (RAG) Chatbot 2026
Personal Project

- Built a from-scratch dense RAG chatbot implementing the full pipeline: PDF chunking, BAAI/bge-base-en-v1.5 embeddings, PostgreSQL + pgvector (HNSW) similarity search, and streaming generation via the Groq LLM API.
- Designed a spoiler-safe retrieval filter and character alias expansion to improve retrieval recall across a multi-book series; built a FastAPI + Pydantic v2 backend with SSE streaming endpoints.

Tech: Python, FastAPI, PostgreSQL, pgvector, sentence-transformers, Groq LLM API, Next.js, TypeScript, RAG

WORK EXPERIENCE

Tri Tech Technology LLC | Full-Stack Developer Intern Jul 2025 – Dec 2025
On-site | Lahore, Pakistan

- Architected and built a 4-service multi-tenant restaurant POS SaaS ecosystem (Node.js/Express API plus three Next.js/TypeScript frontends) sharing one MongoDB-backed REST API.
- Designed multi-tenant data isolation, JWT auth across platform/tenant/PIN login flows, and role-based middleware for five staff roles.
- Deployed the system for real-world restaurant clients with PM2-managed production processes, following Agile practices.

EDUCATION

Bachelor of Science in Computer Science Sep 2022 – Jul 2026
University of Central Punjab (UCP), Lahore | CGPA: 3.60 / 4.00 | Graduated

Relevant Coursework: OOP, DSA, Database Systems, Machine Learning, Deep Learning/AI, NLP, Software Engineering

CERTIFICATIONS

AI Fundamentals, Google/Coursera (Apr 25, 2026) | Fundamentals of Agents: Unit 1, HuggingFace AI Agents Course (May 2, 2026) | Python (Basic), HackerRank Certified (Aug 8, 2026)